NATIONAL LEAGUE OF CITIES

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Urban/Rural Regression Update

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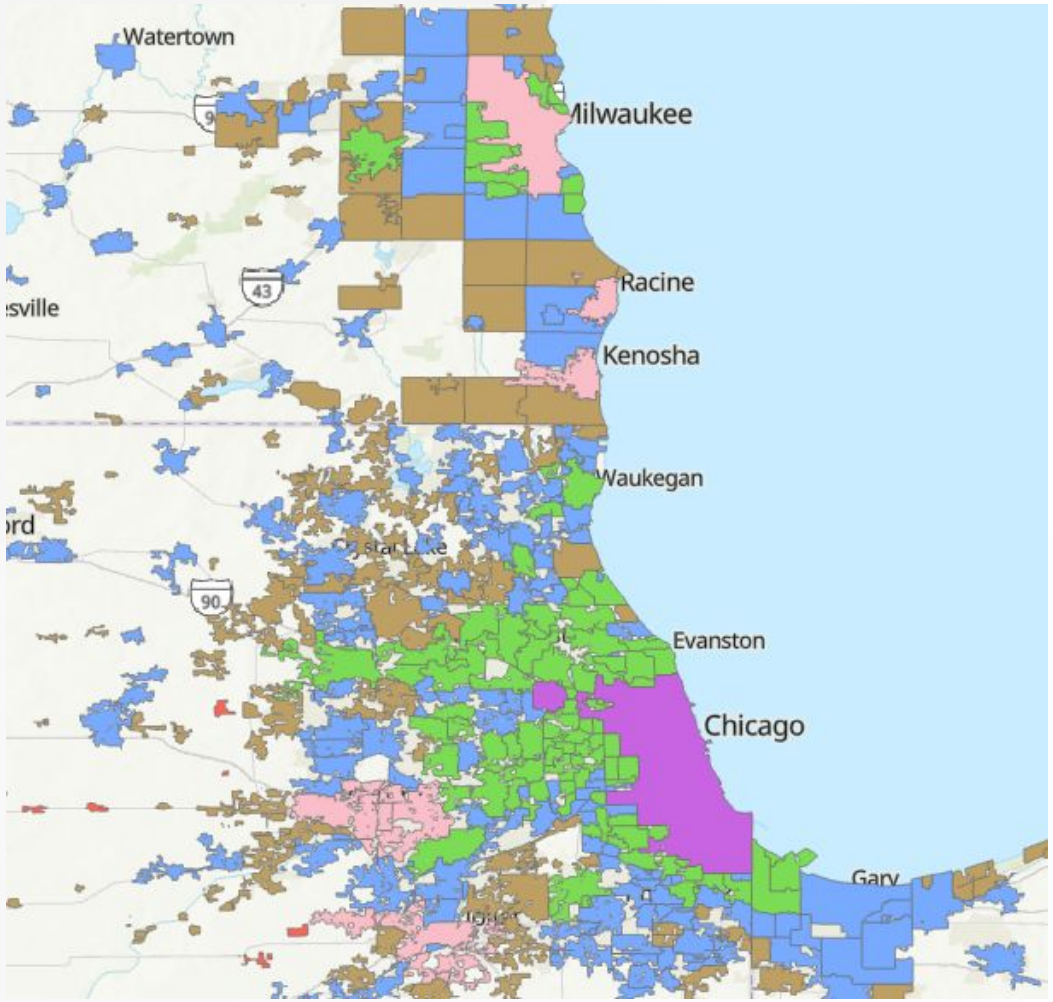
Classifying NLC Members Cities Beyond Urban/Rural Types

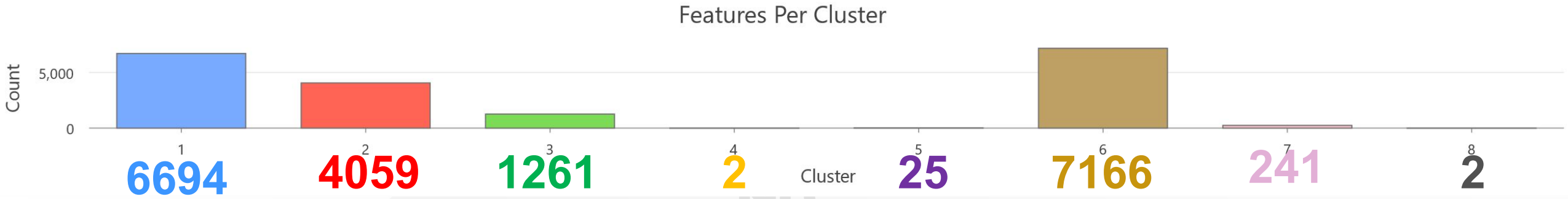
Objective: This brownbag session aims to update NLC on the Urban/Rural Divide project where we use a clustering model to identify American cities by various factors. This classification will help NLC understand how to approach their members' respective needs.

Goal: By the end of this session, NLC staff will have a broader understanding of how this project has developed and open a discussion regarding any concerns/questions on the future of this project.

Results

- While the colors are not directly labeled, we can use data and intuition to determine how the clustering model categorizes over 19,000 cities across the US. This snapshot from ArcGIS image shows similar trends in clusters among two major metro areas Chicago and Milwaukee.





Important Variables-(Economic, Population, Societal Trends)

- Population increase/decrease (% during day)
- Average Number of Jobs (30 min radius), Retail Job Density
- Percentage of property by type (agricultural, industrial..)
- Summed total Assisted & Rented Housing Units(%)
- Frequency of Transit Service During 4-7pm Weekday Peak
- Owner Occupied Home Value of $200,000 to $299,999 (%)

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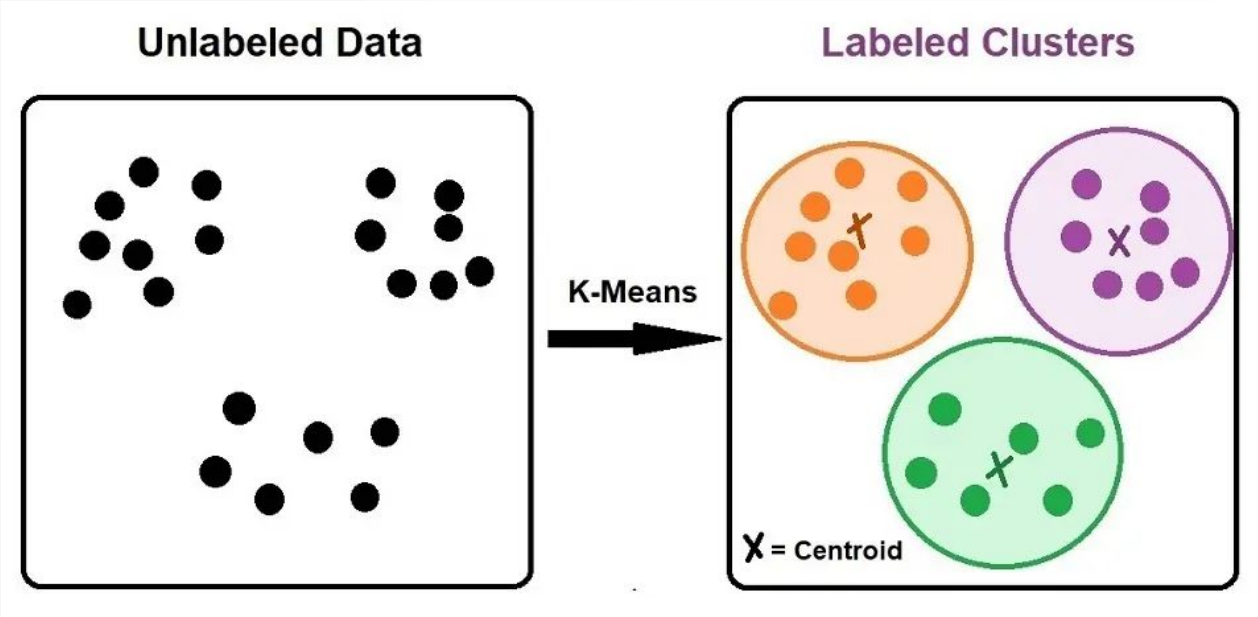
K-means Clustering Algorithm(ArcGIS)

- First decide how many clusters are created. Then the algorithm represents that amount in “center points” and places them randomly near data points with similar data. This process repeats until all data per cluster is averaged out to the point where the center point does not move much at all.
- Disadvantages
  - No initial clear guide into how many groups should be formed, does not reflect the national division.
  - Very sensitive to outliers- outliers may skew the groupings
  - May ignore key differences and focus on certain variables



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Classifying the groups

Pink/Purple Major Metropolitan Area

- Purple New York City, Chicago, LA
- Pink Baltimore, Cleveland, Kansas City

Green/Blue- 1st/2nd Tier Suburb

- Green Chevy Chase, Plano (TX), Burbank (CA)
- Blue- Manassas Park (VA), Laredo (TX), Beford Park

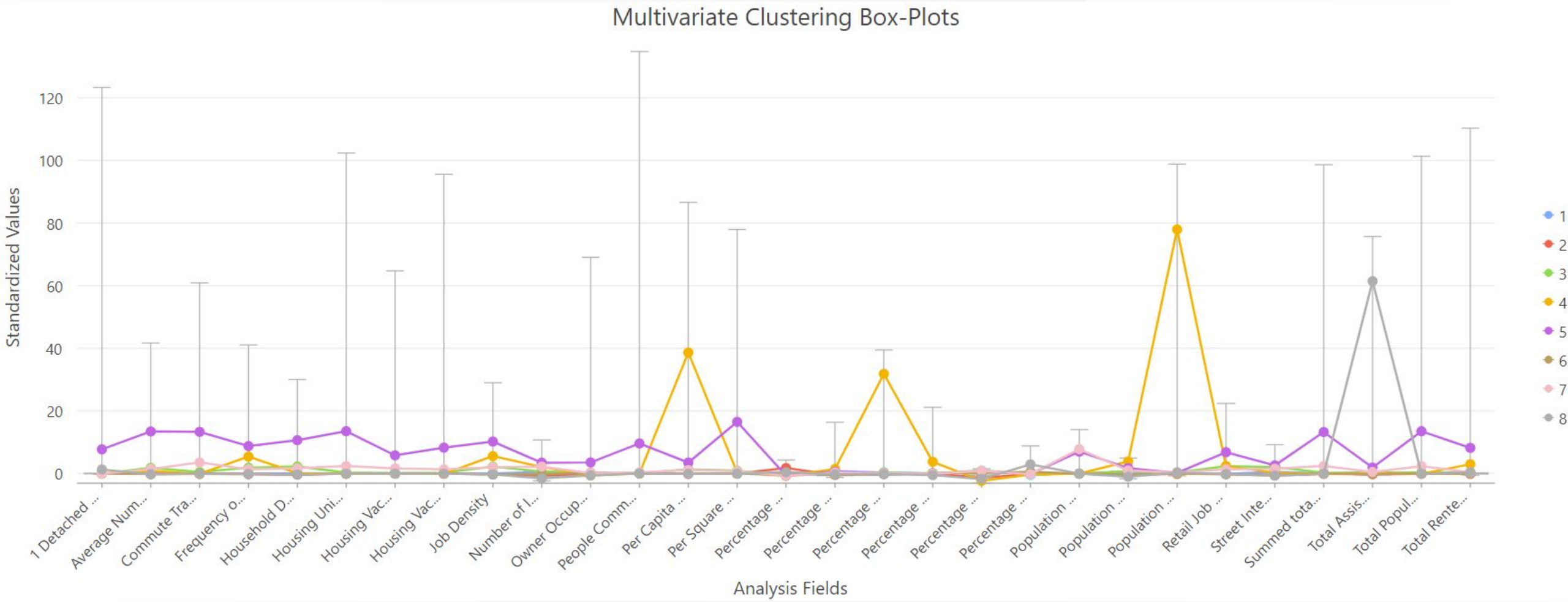
Brown Exurb/Rural

- Poolesville, Malibu (CA), Benson (AZ)

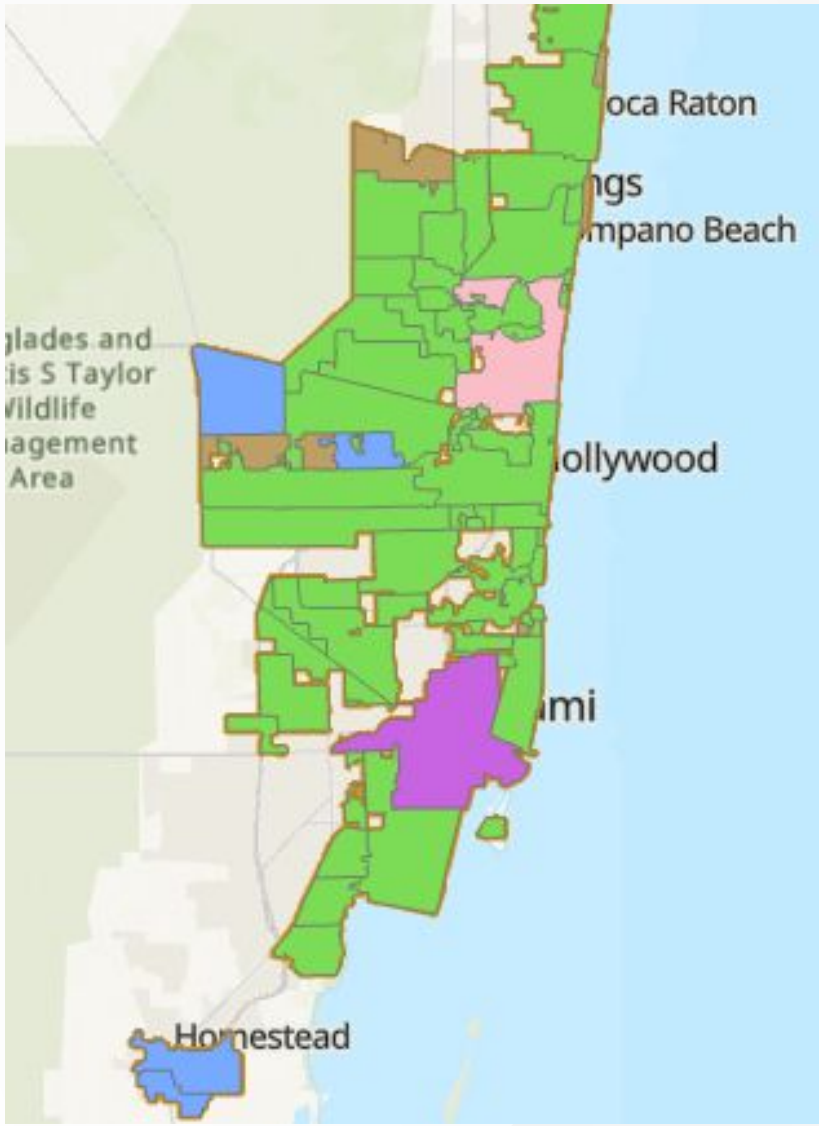
Red- Rural

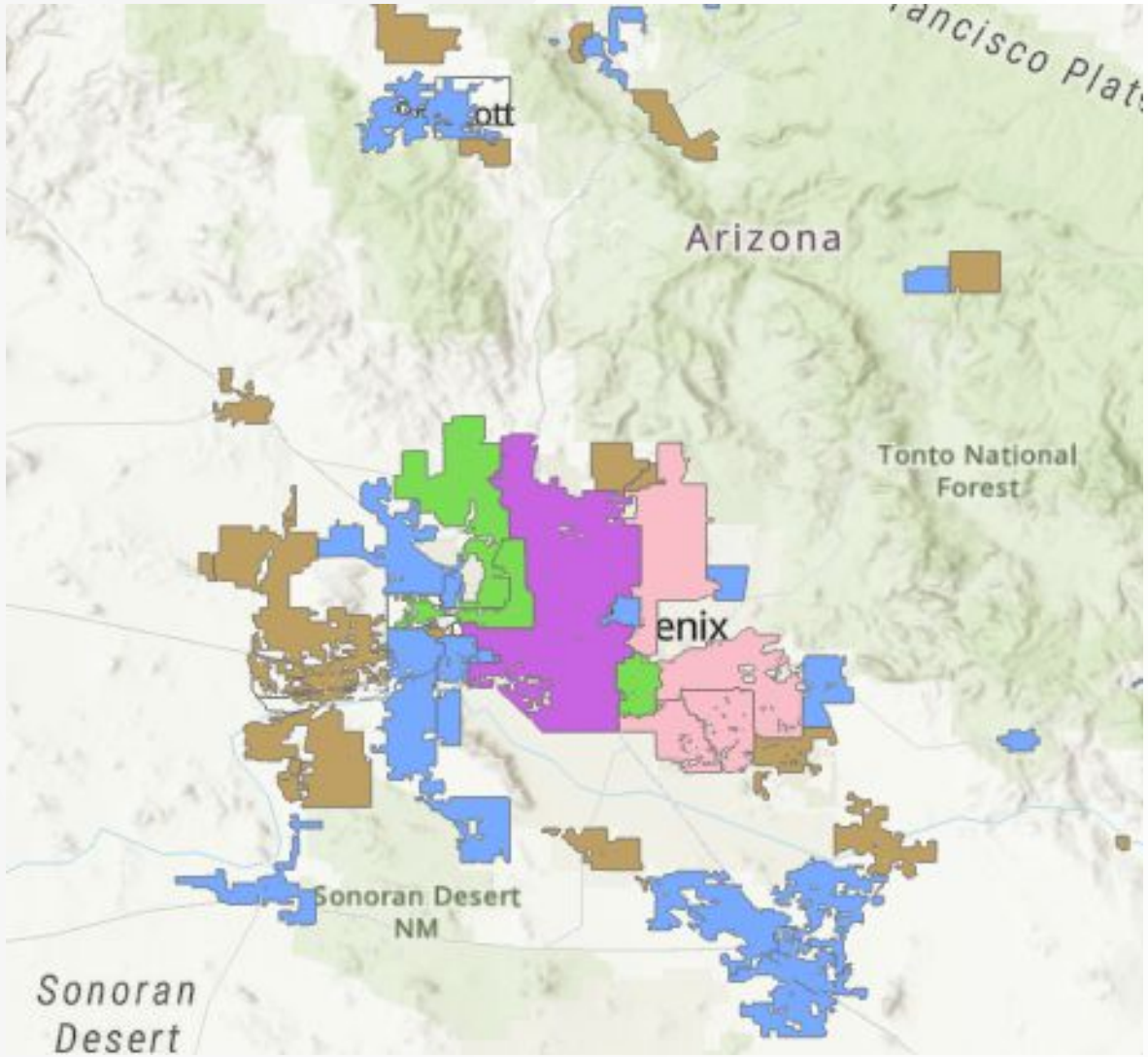
- Loma (ND), Williamsport (OH)
- Highest % of agricultural properties, lowest % of residential properties

8 Cluster Box Plot



Reminder





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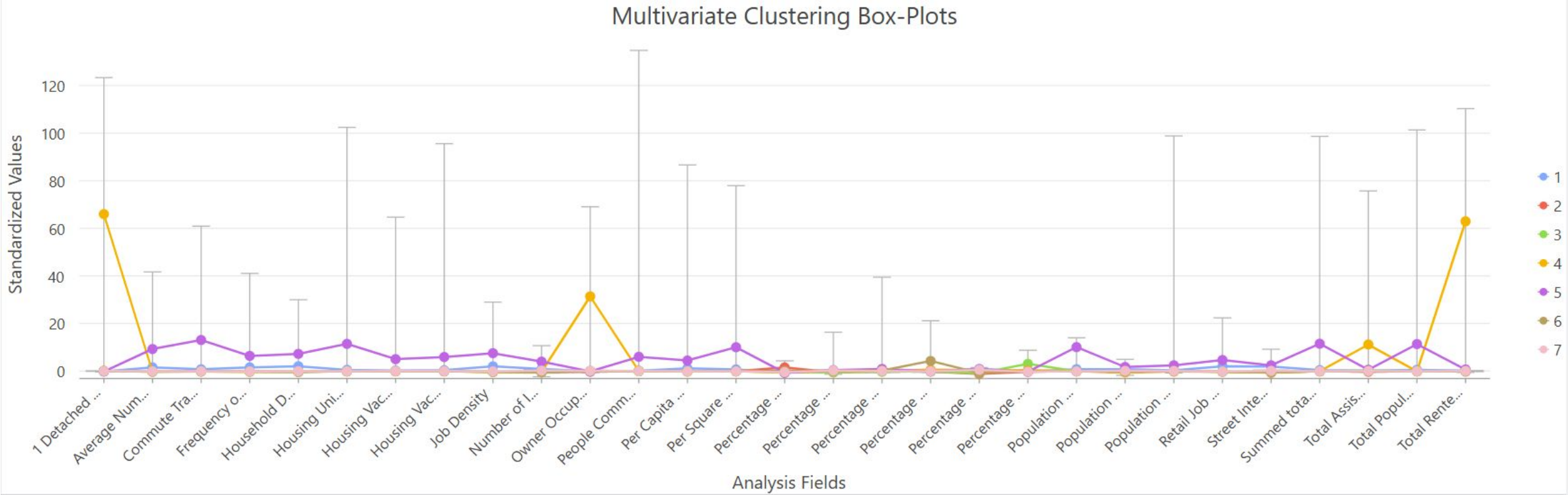
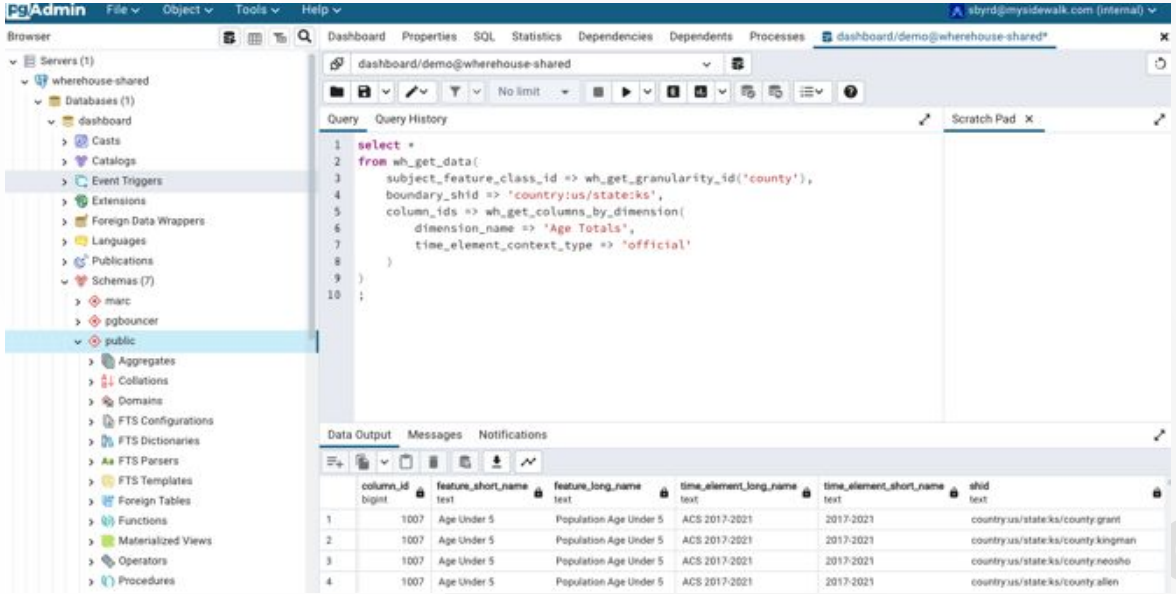
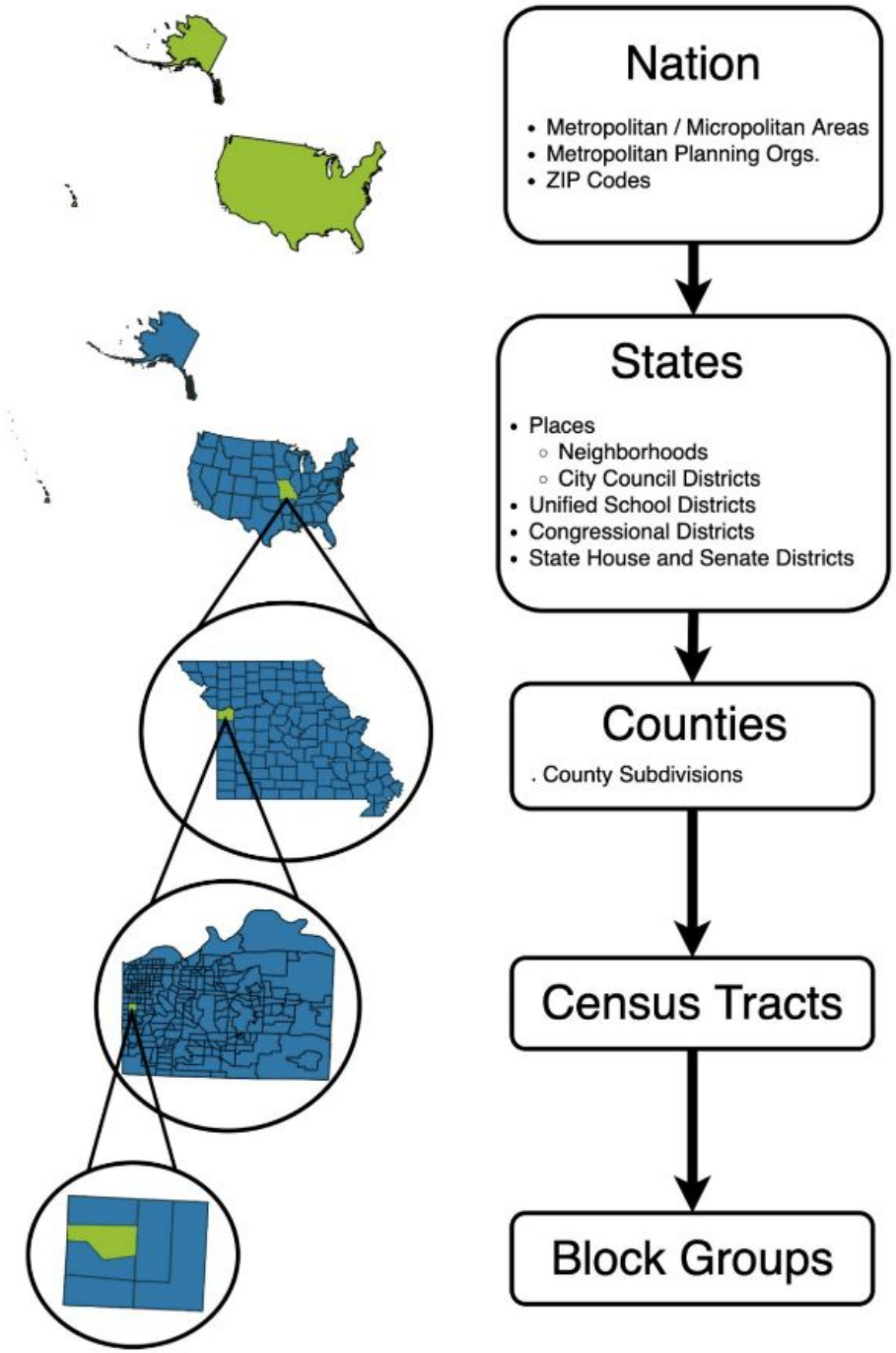
7 Cluster Results



Steps 1-3

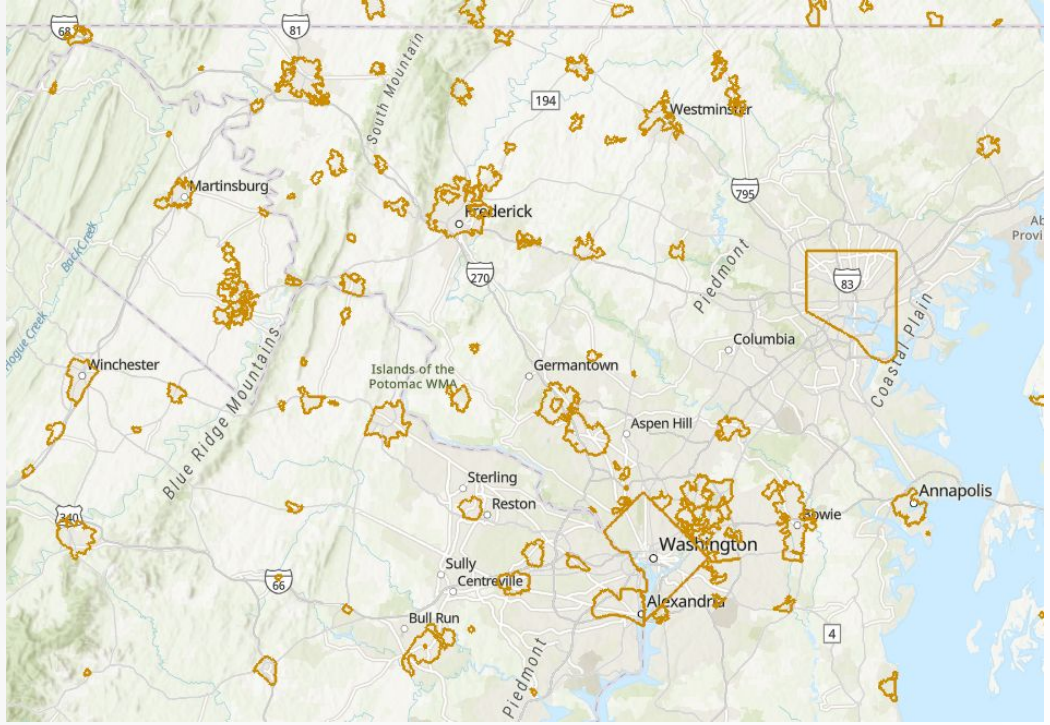
- All data collected were based on Sidewalk “Places” (incorporated cities or Census Designated Places), SQL join was performed to collectively combine “places” data across the US.
- 29 variables (as of now) were chosen

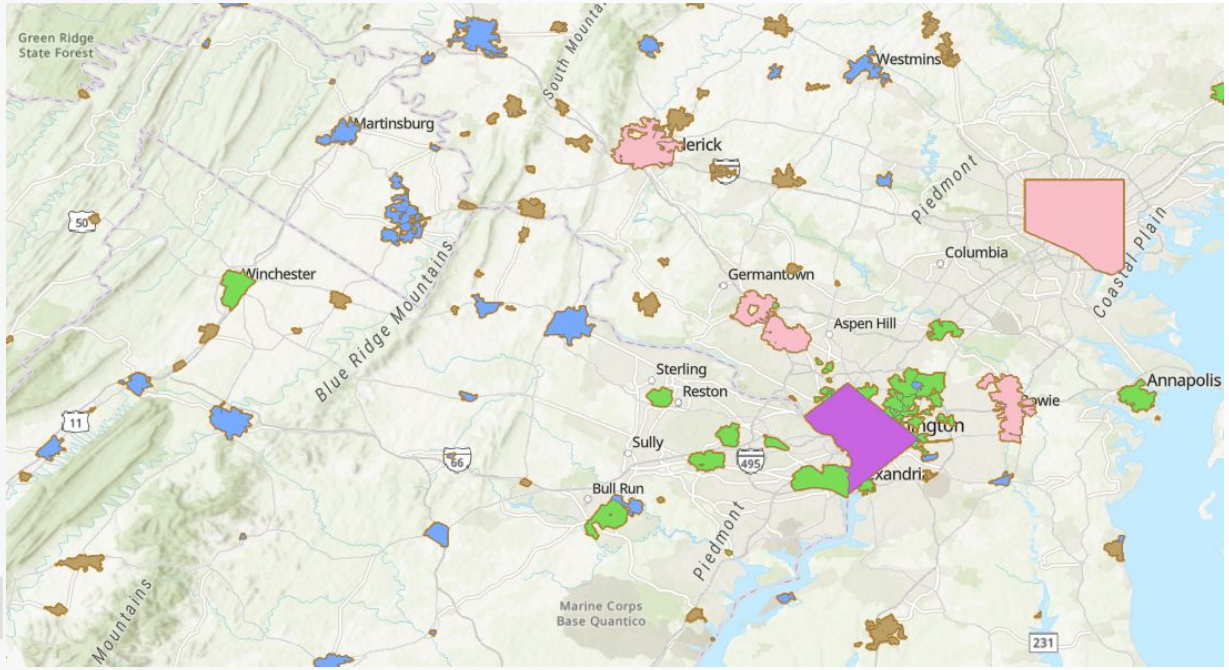




Steps 4-5, ArcGIS import

- Defined Sidewalks places data as “Incorporated Places” in ArcGIS due lack of GeoJSON export, joined with the 28 variables
- Regression results in the discovery of 281 missing places according to ArcGIS, most of which are places from American Samoa and Northern Mariana Islands





Moving Forward

- Considering eliminating/adding new variables based on correlation and diversity needs
  - Suggestions include adding some trend variables, to help distinct unique categories such as exurbs
- Trying new clustering algorithms
- QA/QC, consult experts to determine categories

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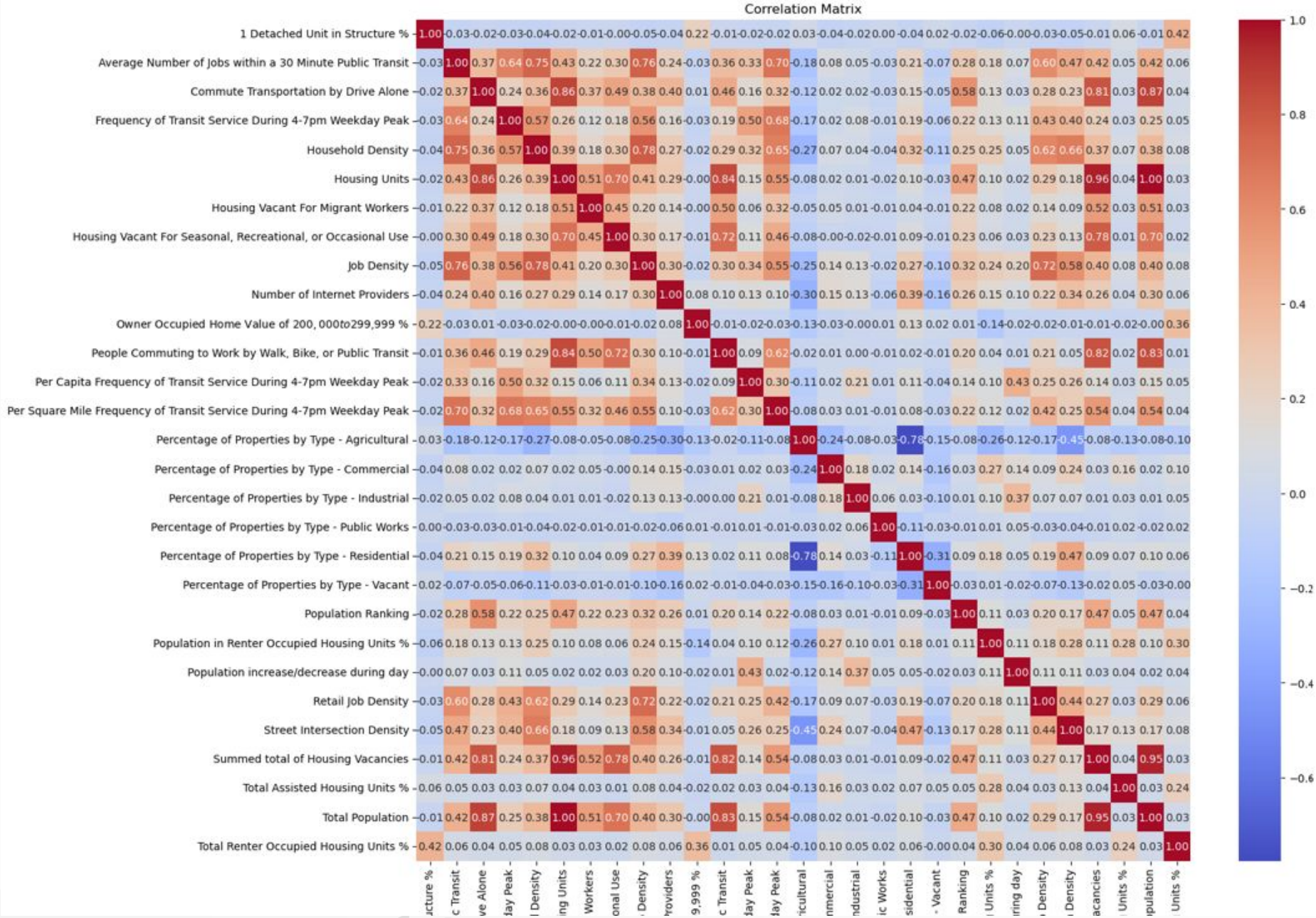
Exploring Variable correlation

Variable pairs with the most correlation-

- Total Population & Housing Units-0.9976
- Housing Units & Summed total of Housing Vacancies- 0.964
- Total Population & Summed total of Housing Vacancies-0.954
- Total Population & Commute Transportation by Drive Alone- 0.874



Process:

1. Identifying variables
2. Attain data from MySidewalk
3. Clean data (such as joins and percentage calculations)
4. Import data into ArcGIS and join with incorporated places
5. Analyze cluster regression results

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Full list of variables:

- 1 Detached Unit in Structure %
- Average Number of Jobs within a 30 Minute Public Transit
- Commute Transportation by Drive Alone
- Frequency of Transit Service During 4-7pm Weekday Peak
- Household Density
- Housing Units
- Housing Vacant For Migrant Workers
- Housing Vacant For Seasonal, Recreational, or Occasional Use
- Job Density\
- Number of Internet Providers
- Owner Occupied Home Value of $200,000 to $299,999 %
- People Commuting to Work by Walk, Bike, or Public Transit
- Per Capita Frequency of Transit Service During 4-7pm Weekday Peak
- Per Square Mile Frequency of Transit Service During 4-7pm Weekday Peak
- Percentage of Properties by Type – Agriculture
- Percentage of Properties by Type – Commercial
- Percentage of Properties by Type – Industrial
- Percentage of Properties by Type - Public Works
- Percentage of Properties by Type – Residential
- Percentage of Properties by Type – Vacant
- Population Ranking
- Population in Renter Occupied Housing Units %
- Population increase/decrease during day
- Retail Job Density
- Street Intersection Density
- Summed total of Housing Vacancies
- Total Assisted Housing Units %
- Total Population
- Total Renter Occupied Housing Units %

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